

The Computer Networks

Lab 1

Creating simple LANs under Packet Tracer

Exercise 1

The purpose of this exercise is to make some simple examples and understand the simulation mode of Packet Tracer.

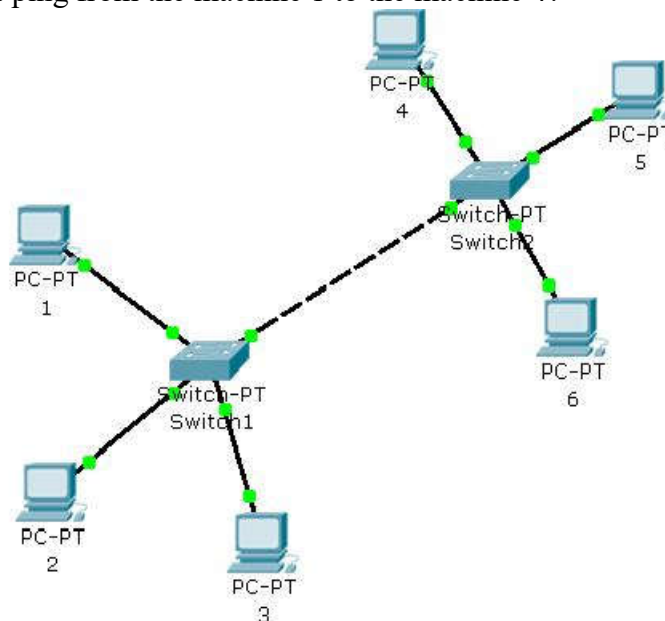
Perform the following edits by configuring each machine (IP address and subnet mask) and for each installation, perform the ping between machines.

1. Connect two computers directly
2. Connect 4 computers via a hub
3. Connect 4 computers via a switch

Exercise 2

Make the example below under Packet Tracer.

1. How many frames are exchanged during a ping from the machine 1 to the machine 2?
(We are interested only in frames containing the ICMP)
2. And during a ping from the machine 1 to the machine 4?



Appendices

I. Straight or crossover Cables

Besides the fact that the new active material ranges automatically adapt to cable, use either crossover or straight cable depending on the type of equipment that is connected:

Straight Cables:

- PC Hub
- PC Switch
- Switch Router

Crossover Cables:

- Switch to Switch
- Hub to Hub
- Router Router
- PC to PC
- Hub Switch
- PC Router

II. PING

The PING command is used to test connectivity to a computer. When doing a ping, it sends a packet of data to a PC and waits for the return.

It launches into the command prompt (command prompt).

Example:

We are on the machine M1 IP = 192.168.1.1 and we want to test the connection with the machine M2 IP = 192.168.1.4

On the machine M, write: ping 192.168.1.4

If all is ok you should have 4 responses and therefore Sent = 4, Received = 4, Lost = 0 (0% loss).

Syntax ping-address or ping PC_name

```
C: \ping gmail.com
```

```
Sending a query 'ping' on gmail.com [74.125.79.83] with 32 bytes of data:
```

```
Response 74.125.79.83: bytes = 32 time = 692 ms TTL = 46
```

```
Response 74.125.79.83: bytes = 32 time = 615 ms TTL = 46
```

```
Response 74.125.79.83: bytes = 32 time = 533 ms TTL = 46
```

```
Response 74.125.79.83: bytes = 32 time = 455 ms TTL = 46
```

```
Ping statistics for 74.125.79.83:
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round in milliseconds:
```

```
Minimum = 455ms, Maximum = 692ms, Average = 573ms
```